

Hip Adduction and Abduction Strength Profiles in Elite Soccer Players

Implications for Clinical Evaluation of Hip Adductor Muscle Recovery After Injury

Kristian Thorborg,^{*†} M. Sportsphysio., Andreas Serner,[‡] BPthy, Jesper Petersen,[†] PhD, Thomas Moller Madsen,[‡] BPthy, Peter Magnusson,[§] DSc, and Per Hölmich,[†] MD
Investigation performed at the Department of Orthopaedic Surgery, Amager Hospital, University of Copenhagen, Denmark

Background: An ipsilateral hip adduction/abduction strength ratio of more than 90%, and hip adduction strength equal to that of the contralateral side have been suggested to clinically represent adequate strength recovery of hip adduction strength in athletes after groin injury. However, to what extent side-to-side symmetry in isometric hip adduction and abduction strength can be assumed in soccer players remains uncertain.

Purpose: To compare isometric hip adduction and abduction strength on the dominant and nondominant side in injury-free soccer players.

Study Design: Cross-sectional study; Level of evidence, 3.

Methods: One hundred elite soccer players were included. Maximal unilateral isometric hip adduction and abduction strength on the dominant and nondominant side were measured with a handheld dynamometer, using a reliable test procedure.

Results: The dominant side was stronger than the nondominant side for both isometric hip adduction (2.45 ± 0.54 vs 2.37 ± 0.48 Nm/kg, $P = .02$) and hip abduction (2.35 ± 0.33 vs 2.25 ± 0.31 Nm/kg, $P < .001$), corresponding to a 3% and 4% difference, respectively. Isometric hip adduction was greater than isometric hip abduction for both the dominant (2.44 ± 0.53 vs 2.35 ± 0.33 Nm/kg, $P = .04$) and nondominant (2.37 ± 0.48 vs 2.26 ± 0.33 Nm/kg, $P = .03$) side. Isometric hip adduction/abduction ratio was not different between the dominant (1.04 ± 0.18) and nondominant (1.06 ± 0.17 , $P = .40$) side. A post hoc analysis showed that isometric hip adduction/abduction ratio was significantly lower in players with groin pain during hip adduction testing compared with players with a pain-free test (0.80 ± 0.14 , $P < .001$).

Conclusion: The marginal difference between the dominant and the nondominant side is within the measurement variation of the test procedure, and contralateral isometric hip adduction strength can therefore be used as a simple clinical reference point of full recovery of hip adduction muscle strength in soccer players. Furthermore, it is suggested that the ipsilateral hip adduction/abduction strength ratio is used as a guideline for evaluating hip adduction strength recovery in soccer players with bilateral groin problems.

Keywords: hip adduction strength; limb dominance; hip adduction/abduction ratio; groin pain; soccer players; handheld dynamometer

*Address correspondence to Kristian Thorborg, MSc, Department of Orthopaedic Surgery, Amager Hospital, Italiensvej 1, DK-2300 Copenhagen, Denmark (e-mail: kristianthorborg@hotmail.com).

[†]Department of Orthopaedic Surgery, Amager Hospital, Faculty of Health Sciences, University of Copenhagen, Denmark.

[‡]Faculty of Physiotherapy, Metropolitan University College, Copenhagen, Denmark.

[§]Institute of Sports Medicine Copenhagen, Bispebjerg Hospital, University of Copenhagen, Denmark.

The authors declared that they had no conflicts of interest in their authorship and publication of this contribution.

Isometric hip strength plays an important role in the clinical assessment of soccer players with hip and groin injuries.^{11,12} Generally it has been suggested that lower extremity strength deficits of less than 10% on the injured side compared with the uninjured side should be considered the clinical milestone before returning an athlete to sport after an injury.^{1,19} More specifically, the achievement of a hip adduction/abduction ratio of more than 90% and an adduction strength equal to that of the contralateral side has been recommended before returning to sport after an adductor strain.¹⁸

A recent study²¹ has shown that eccentric hip adduction symmetry cannot be assumed in injury-free soccer players.

In fact, the dominant side was 14% stronger than the non-dominant side with regard to eccentric hip adduction, although hip abduction strength was similar.²¹ This finding of asymmetric eccentric hip adduction strength in injury-free soccer players between the dominant and non-dominant side indicates that using contralateral eccentric hip adduction strength as a reference point for muscle recovery is questionable.

Clinically, isometric tests are often preferred.^{10,11,24} Isometric testing has the advantage of producing less stress on the musculoskeletal system than eccentric testing,³ and thus minimizing the risk of injury and delayed-onset muscle soreness.^{7,8} However, whether an isometric hip abduction and hip adduction strength symmetry in injury-free soccer players can be assumed is uncertain because eccentric hip adduction strength asymmetries have been shown in soccer players.²¹

The purpose of this study was to compare isometric hip adduction and hip abduction strength on the dominant and the nondominant kicking side in injury-free soccer players.

MATERIALS AND METHODS

One hundred male elite soccer players from 6 different clubs gave informed consent, in accordance with the Helsinki Declaration, to participate in the study. All players (age, 24 ± 4 years [mean $\pm$ SD]; height, 183 ± 7 cm; weight, 79 ± 7 kg) played at the third best level (division 2, East region) in the national competition in Denmark. The players were semiprofessional; they trained 3 to 4 times a week and played 1 to 2 games weekly.

The players were tested in February 2009 during the midseason competition break leading up to the second half of the season, which started approximately 1 month later. Injured players were not included. All players were tested when they attended a training session (before training), and only players who participated in the training session were included in the study. No exercise/training or match was allowed on the day before the test. Players were excluded if they were unable to participate in the training as a result of injury at the time of testing or if they were taking any pain medication. A self-reported questionnaire was given to the subjects before testing to determine limb dominance and injury history. The preferred lower limb for kicking a ball was defined as the dominant side. Injury history, restricted to hip and/or groin pain within the previous year, was recorded to avoid recall bias related to the collection of retrospective injury data from several body regions.⁹ The self-reported questionnaire was administered by a physiotherapist (T.M.M.) who was not involved in the strength testing.

Testing Procedures

The testing setup included a portable handheld dynamometer (PowerTrack II Commander, JTECH Medical, Salt Lake City, Utah) and an examination table. The handheld dynamometer was calibrated before testing and all test procedures were standardized. Maximal voluntary

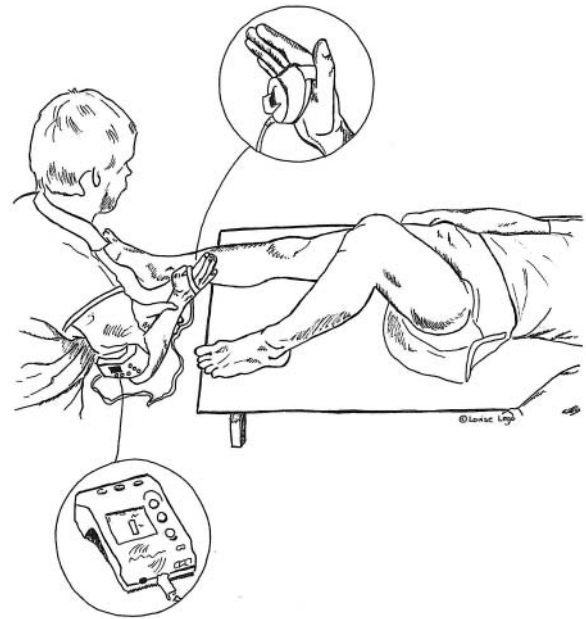


Figure 1. Testing setup for “make” test. Illustration from Thorborg et al,²² with permission from *Scandinavian Journal of Medicine and Science in Sports* (2010, John Wiley & Sons A/S).

isometric hip adduction and hip abduction torque for the dominant and nondominant side were tested for all the participants. The testing sequence was randomized before the testing sessions to avoid systematic bias. Isometric hip adduction and hip abduction were measured in a “make” test in the supine position as introduced by Thorborg et al²² (Figure 1), and calculation of the hip adduction/abduction ratio was based on these measurements. Substantial intratester, interday reliability of the isometric strength testing and the hip abduction/adduction ratio has previously been reported.²²

The participants were placed in the supine position and were told to stabilize themselves by holding onto the side of the table with their hands. The examiner, a physiotherapist (A.S.), applied resistance in a fixed position 5 cm proximal to the proximal edge of the lateral malleolus, and the person being tested exerted a 5-second maximum isometric voluntary contraction against the dynamometer. After the participants were instructed in the procedures, they were asked to perform one practice trial. The highest of the 4 subsequent measurements were used in the analysis. If the last measurement was the highest, another measurement was conducted until no further force increase was measured. There was a 30-second rest period between trials. The rest period was introduced to avoid a decline in strength across trials due to fatigue.²⁰ The standardized command by the examiner was “go ahead: push-push-push-push-relax.” The same examiner (A.S.), who was blinded to side dominance and pain during testing, performed all the measurements.

After each strength test trial, players were asked to rate any pain during testing on a numerical visual analog scale (VAS) and this was recorded. The players reporting pain during strength testing were asked to specify the location

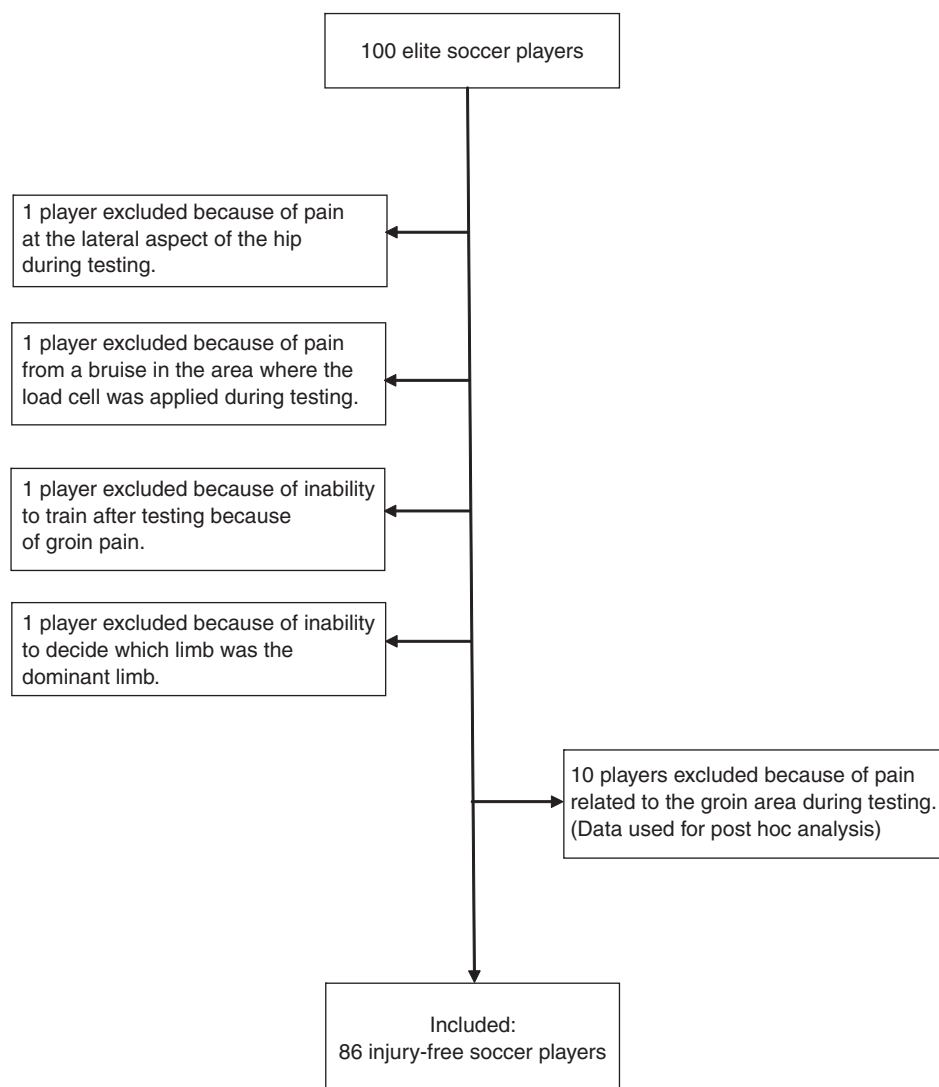


Figure 2. Reasons for exclusion of players from hip torque analysis.

of the pain, and this was also recorded. Data from players reporting pain during testing were excluded from the analysis of isometric hip strength as pain has been shown to reduce force production.²

Limb length was measured from the most prominent point on the anterior superior iliac spine in the supine position to 5 cm proximal to the proximal edge of the lateral malleolus. Limb length was used to calculate torque, and all force values were weight adjusted and reported as Newton-meters per kilogram of body weight.

Ethical approval was obtained from The Danish Ethics Committee of the Capital Region and the Danish Data Protection Agency.

Statistical Analysis and Data Reduction

Data are presented as mean $\pm$ 1 standard deviation (SD). All data were statistically examined for normality of

distribution (Kolmogorov-Smirnov). The nondominant hip abduction data set was the only data set not being normally distributed; all data in the analyses, including the nondominant hip abduction data set, were therefore log-transformed. Furthermore, 2-sided F test was not significant for any unpaired comparisons; thus parametric statistics were applied. Paired and unpaired Student t tests were used appropriately. A level of $P < .05$ was chosen to indicate statistical significance. The Grubb test was used to detect outliers in the individual test, which were removed (www.graphpad.com, 2009).

Data from 14 players were excluded from the analysis of isometric hip adduction and hip abduction torque for the following reasons: 1 player reported lateral hip pain during testing, 1 player reported pain from a bruise in the area where the load cell was applied during testing, 1 player had no preferred dominant limb, 1 player was not able to train because of groin pain, and 10 players reported groin pain during hip adduction testing corresponding to the area of the adductor

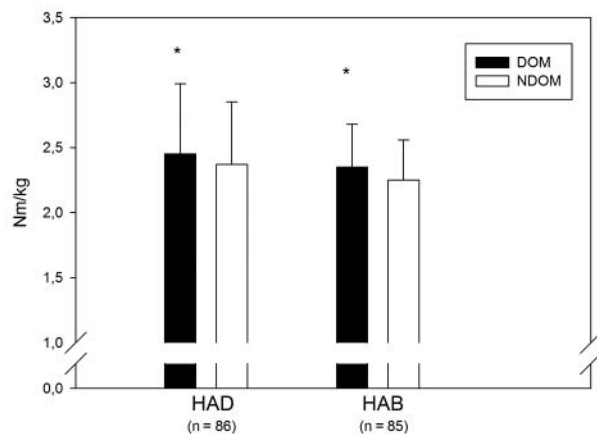


Figure 3. Hip adduction and abduction torque profiles on the dominant (DOM) and nondominant (NDOM) side. HAD, hip adduction; HAB, hip abduction. *Statistically significant ($P < .05$) within-group difference between DOM and NDOM side for HAD and HAB, respectively.

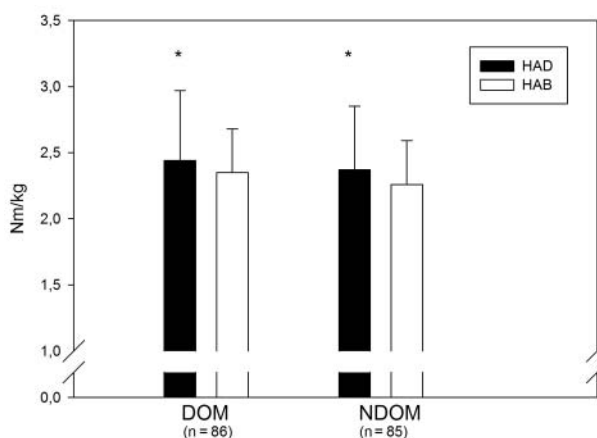


Figure 4. Hip adduction (HAD) versus hip abduction (HAB) strength profiles in soccer players. DOM, dominant side; NDOM, nondominant side. *Statistically significant ($P < .05$) within-group difference between HAD and HAB for DOM and NDOM side, respectively.

muscles' proximal tendons, insertion, or the pubic symphysis. It was decided to use the data for the 10 players in a post hoc analysis of hip torque comparing players with groin pain during hip adduction testing with players with a pain-free test to investigate whether groin pain (VAS, 3.9 ± 2.1) during hip adduction testing has any implications for hip adduction/abduction ratio values (Figure 2).

All data were checked for outliers. One outlier in the group of players with a pain-free test was detected in the dominant hip abduction data set. This outlier was removed because the test value indicated that an error had occurred in the measurement procedure or the data collection during the dominant hip abduction testing concerning this subject.

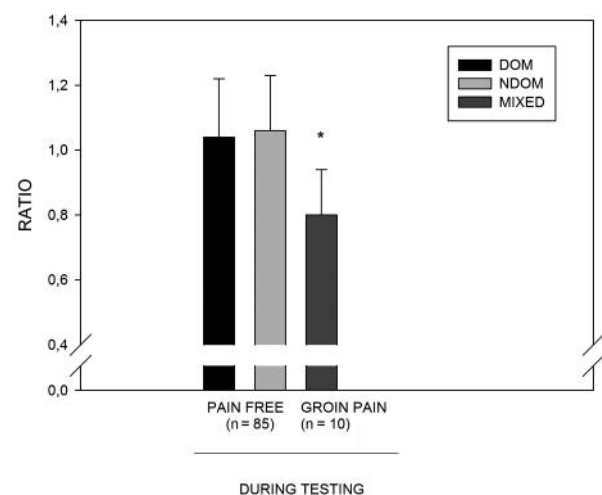


Figure 5. Hip adduction/abduction ratio. DOM, dominant side; NDOM, nondominant side. *Statistically significant ($P < .05$) between-group difference between MIXED, and DOM and NDOM side, respectively.

RESULTS

Of the 86 players included in this study, 69% reported that they had experienced hip and/or groin pain during the previous year. Of the 10 players with groin pain during testing, 80% reported that they had experienced hip and/or groin pain during the previous year.

Hip torque values are shown in Figures 3 and 4. The dominant side was stronger than the nondominant side for both isometric hip adduction (2.45 ± 0.54 vs 2.37 ± 0.48 N·m/kg, $P = .02$) and hip abduction (2.35 ± 0.33 vs 2.25 ± 0.31 N·m/kg, $P < .001$), corresponding to a 3% and 4% difference, respectively. Isometric hip adduction was greater than isometric hip abduction for both dominant (2.44 ± 0.53 vs 2.35 ± 0.33 N·m/kg, $P = .04$) and nondominant (2.37 ± 0.48 vs 2.26 ± 0.33 N·m/kg, $P = .03$) side. Isometric hip adduction/abduction ratio was not different between the dominant (1.04 ± 0.18) and nondominant (1.06 ± 0.17 , $P = .40$) side. A post hoc analysis showed that isometric hip adduction/abduction ratio was significantly lower in players with groin pain during hip adduction testing compared with players with a pain-free test (0.80 ± 0.14 , $P < .001$) (Figure 5).

DISCUSSION

To our knowledge, the present study is the first investigation of isometric hip adduction and hip abduction strength in elite soccer players using a unilateral strength test. The data show that isometric hip adduction and abduction strength is only marginally greater on the dominant than the nondominant side in soccer players. A difference of 3% to 4% in isometric hip adduction and hip abduction strength difference in soccer players is small, especially compared

with eccentric hip adduction strength testing in which the difference between the dominant and the nondominant side can reach up to 14%.²¹ The reason for the seemingly larger difference between dominant and nondominant hip adduction strength during eccentric testing compared with isometric testing is unknown. However, because eccentric EMG activity in the hip adductors is substantial during the early swing phase of a soccer kick,^{4,5} it is possible that an improved eccentric hip adduction strength adaption on the dominant side is a result of improved coordination of the motor system from repetitive kicking.

An intratester, interday reliability study has shown that the absolute measurement variation for the testing procedure used in the present study is 6% for isometric hip adduction testing and 3% for isometric hip abduction.²² This indicates that a 3% hip adduction strength difference between the dominant and nondominant side would not be possible to measure at the individual level. A 3% difference in hip adduction torque between the dominant and nondominant side therefore does not seem to be of any clinical relevance, and isometric hip adduction strength of the dominant and nondominant side should be considered equal for the individual. This has important clinical implications as the strength of the contralateral side can be used as a relevant reference point of full recovery of hip adduction strength. A study from 1988 investigating isometric hip abduction torque in 40 healthy, non-soccer-playing subjects of similar age using a similar test procedure showed a 10% difference between the dominant and nondominant side.¹⁷ The present study shows that soccer players do not seem to have a similar substantial isometric hip abduction torque difference between the dominant and nondominant side, which might be explained by the fact that soccer induces substantial hip abductor muscle activity on the nondominant (stabilizing) side during kicking.⁴ A recent study investigating eccentric hip abduction torque was unable to show any difference between the dominant and nondominant side; however, this may be related to the small sample size ($n = 9$) of the study, which increases the likelihood of a type II error.²¹

A study from 1968 investigating isometric hip adduction and abduction torque in 80 healthy, non-soccer-playing subjects, of similar age and using a similar test procedure, showed a 7% greater hip adduction strength compared with hip abduction strength,¹⁶ which is in accordance with the results of the present study. Our results imply that the isometric hip adduction/abduction ratio is approximately 1.05 in soccer players irrespective of which side is tested. Clinically, this finding has implications because it indicates that the ipsilateral hip adduction/abduction ratio can be used as a guideline during evaluation of recovery of hip adduction strength in soccer players with groin injury. The advantage of using the ipsilateral side as a guideline is that current or previous injury to the other leg will not interfere with this measurement procedure. This is relevant because groin pain in athletes is often experienced bilaterally,¹² and using the contralateral leg as a reference point can therefore present a clinical problem.

By testing each leg individually we believe that we achieve a greater depiction of the actual muscle performance

in hip adduction in both the injured and uninjured leg compared with using a squeeze test, in which the produced torque will be a combination of the hip adduction performance of both legs.^{14,15} The squeeze test is performed by using the cuff of a sphygmomanometer, which is placed between the knees, and the athletes are then instructed to squeeze the cuff as hard as they can using both legs. The highest pressure displayed on the sphygmomanometer dial (to the nearest 5 mm Hg) during the test is then recorded.¹⁴ In subjects with pain elicited during the squeeze test, the maximal force production will be determined by the painful side in subjects with unilateral groin pain and by the most painful side in subjects with bilateral groin pain. Therefore, we think it is important to consider these aspects clinically when using a bilateral squeeze test or a unilateral strength tests for evaluation of hip adductor muscle strength and recovery.

Groin pain is often aggravated during important soccer skills such as kicking, accelerating, and sudden change of direction.¹³ A post hoc analysis revealed that the isometric hip adduction/abduction torque ratio is 24% lower in soccer players with groin pain (VAS, 3.9 ± 2.1) during hip adduction testing than in players with a pain-free test. A decreased hip adduction/abduction ratio during maximal exertion might have implications for soccer players and their ability to perform at the maximum level. The percentage of players who experienced hip and/or groin pain during the previous year was almost identical in the 2 groups, indicating that groin pain during hip adduction testing could be a more sensitive indication than injury history of which players are displaying reduced hip muscle function. It seems that groin pain during hip adduction testing negatively affects the ability to produce maximal hip adduction torque and thereby decreases the hip adduction/abduction ratio to approximately 80%. This finding is in accordance with the study by Malliaras et al,¹⁴ who showed that athletes with groin pain displayed an average reduced hip adduction force in the squeeze test of approximately 20% compared with healthy controls. Unfortunately, the authors did not report whether any pain was encountered during testing.

Not recording possible pain during testing seems to be a general tendency in studies finding hip adduction torque reductions in athletes.^{6,10,15,23} Previously it has been shown that reducing pain during testing in the shoulder region increases force production.² Because pain may be a possible confounder when testing muscle strength, we suggest that pain during testing be reported in future studies concerning athletes and hip strength measurements and that the relationship between reported pain and hip adduction strength should be further investigated.

A limitation of the present study is the high proportion of soccer players reporting hip and/or groin pain during the previous year. Soccer players are very exposed to hip and/or groin pain, and about 70% of all players reported hip and/or groin pain during the previous year, which is in accordance with previous reporting.¹⁰ Hip and/or groin pain during the previous year may influence the force values in the population under study; however, because hip and/or groin pain is highly prevalent in soccer players at

this level, we chose to include players who were participating fully in training because we wanted the test to be performed on the population for which it was intended, namely active elite soccer players. Therefore, excluding soccer players because of previous history of hip and/or groin pain would mean that our data could only be extrapolated to a very small and potentially irrelevant subgroup of players. Whether the data from our study can be extrapolated to all elite soccer players is unknown, but an advantage of using the contralateral or ipsilateral side for comparison instead of generalized normative values is that the wide range of individual muscle strength will not be a problem. However, it is unknown whether players who are stronger or weaker than those of the present study display different strength symmetry indexes.

The present testing method is a simple way of testing muscular strength of hip adduction and abduction because it only requires the use of an examination table, a handheld dynamometer, a weighing scale, and a measuring tape. The hip adduction/abduction ratio seems to be a relevant measure of hip strength, especially in soccer players with bilateral groin symptoms. We therefore recommend the use of unilateral tests of hip adduction and abduction strength in future studies regarding soccer players with and without groin pain as a useful outcome measure of hip adductor and abductor muscle function. For clinical evaluation of the individual soccer player, the obvious advantage is that the testing method does not require any age, leg length, or weight adjustment because the player can act as his own control. This makes this testing method ideal for quick assessments in the busy clinical situation.

In conclusion, isometric hip adduction strength was only 3% greater on the dominant than the nondominant side of the players tested. This marginal difference is within the measurement variation of the procedure; therefore, contralateral isometric hip adduction strength is suggested as a simple clinical reference point of full recovery of isometric hip adduction muscle strength in soccer players. Furthermore, we suggest that the ipsilateral hip adduction/abduction strength ratio be used as a guideline for evaluating hip adduction strength recovery in soccer players with bilateral groin problems.

REFERENCES

1. Augustsson J, Thomee R, Karlsson S. Ability of a new hop test to determine functional deficits after anterior cruciate ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 2004;12:350-356.
2. Ben-Yishay A, Zuckerman JD, Gallagher M, Cuomo F. Pain inhibition of shoulder strength in patients with impingement syndrome. *Orthopedics*. 1994;17:685-688.
3. Bohannon RW. Make tests and break tests of elbow flexor muscle strength. *Phys Ther*. 1988;68:193-194.
4. Brophy RH, Backus SI, Pansy BS, Lyman S, Williams RJ. Lower extremity muscle activation and alignment during the soccer instep and side-foot kicks. *J Orthop Sports Phys Ther*. 2007;37:260-268.
5. Charnock BL, Lewis CL, Garrett WE Jr, Queen RM. Adductor longus mechanics during the maximal effort soccer kick. *Sports Biomech*. 2009;8:223-234.
6. Crow JF, Pearce AJ, Veale JP, Vanderwesthuizen D, Coburn PT, Pizzari T. Hip adductor muscle strength is reduced preceding and during the onset of groin pain in elite junior Australian football players. *J Sci Med Sport*. 2010;13:202-204.
7. Friden J, Sfakianos PN, Hargens AR. Muscle soreness and intramuscular fluid pressure: comparison between eccentric and concentric load. *J Appl Physiol*. 1986;61:2175-2179.
8. Friden J, Sjöström M, Ekblom B. Myofibrillar damage following intense eccentric exercise in man. *Int J Sports Med*. 1983;4:170-176.
9. Gabbe BJ, Finch CF, Bennell KL, Wajswelner H. How valid is a self reported 12 month sports injury history? *Br J Sports Med*. 2003;37:545-547.
10. Hanna CM, Fulcher ML, Elley CR, Moyes SA. Normative values of hip strength in adult male association football players assessed by handheld dynamometry. *J Sci Med Sport*. 2010;13:299-303.
11. Holmich P, Holmich LR, Bjerg AM. Clinical examination of athletes with groin pain: an intraobserver and interobserver reliability study. *Br J Sports Med*. 2004;38:446-451.
12. Holmich P, Uhrskou P, Ulnits L, et al. Effectiveness of active physical training as treatment for long-standing adductor-related groin pain in athletes: randomised trial. *Lancet*. 1999;353:439-443.
13. Lynch SA, Renstrom PAFH. Groin injuries in sport: treatment strategies. *Sports Med*. 1999;28:137-144.
14. Malliaras P, Hogan A, Nawrocki A, Crossley K, Schache A. Hip flexibility and strength measures: reliability and association with athletic groin pain. *Br J Sports Med*. 2009;43:739-744.
15. Mens J, Inklaar H, Koes BW, Stam HJ. A new view on adduction-related groin pain. *Clin J Sport Med*. 2006;16:15-19.
16. Murray MP, Sepic SB. Maximum isometric torque of hip abductor and adductor muscles. *Phys Ther*. 1968;48:1327-1335.
17. Neumann DA, Soderberg GL, Cook TM. Comparison of maximal isometric hip abductor muscle torques between hip sides. *Phys Ther*. 1988;68:496-502.
18. Nicholas SJ, Tyler TF. Adductor muscle strains in sport. *Sports Med*. 2002;32:339-344.
19. Orchard J, Best TM, Verrall GM. Return to play following muscle strains. *Clin J Sport Med*. 2005;15:436-441.
20. Sisto SA, Dyson-Hudson T. Dynamometry testing in spinal cord injury. *J Rehabil Res Dev*. 2007;44:123-136.
21. Thorborg K, Couppe C, Petersen J, Magnusson P, Holmich P. Eccentric hip adduction and abduction strength in elite soccer players and matched controls: a cross-sectional study [published online ahead of print May 29, 2010]. *Br J Sports Med*.
22. Thorborg K, Petersen J, Magnusson SP, Holmich P. Clinical assessment of hip strength using a hand-held dynamometer is reliable. *Scand J Med Sci Sports*. 2010;20(3):493-501.
23. Tyler TF, Nicholas SJ, Campbell RJ, McHugh MP. The association of hip strength and flexibility with the incidence of adductor muscle strains in professional ice hockey players. *Am J Sports Med*. 2001;29:124-128.
24. Verrall GM, Slavotinek JP, Barnes PG, Fon GT. Description of pain provocation tests used for the diagnosis of sports-related chronic groin pain: relationship of tests to defined clinical (pain and tenderness) and MRI (pubic bone marrow oedema) criteria. *Scand J Med Sci Sports*. 2005;15:36-42.